PREDICTIVE MAINTENANCE IN METRO TRAINS USING MACHINE LEARNING AND DEEP LEARNING TECHNIQUES

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AST Project Work — Course Project Review 3 Final Capstone Report

ABSTRACT

(1) Project Overview: This project designs, implements, and benchmarks an industrial-grade predictive maintenance framework for Air Production Units (APU) in passenger metro trains operated by *Metro do Porto, Portugal*. By continuously monitoring 16 multivariate analog and digital sensor telemetry channels, the system anticipates impending pneumatic compressor leaks and lubrication failures up to two hours before physical system breakdown occurs.

(2) Base Paper Algorithms & Their Accuracy: Base literature (Matthias Jakobs et al., 2026 [1]; Bruno Veloso et al., 2022 [16]) investigated online rule induction (RIPPER, Decision Trees) and classical tree ensembles (Random Forest, XGBoost) on MetroPT telemetry. These reported an initial failure detection F1-score of ~0.65 to 0.78, but suffered from severe false alarm rates (precision < 50%) under extreme streaming class imbalance and exhibited high vulnerability to sensor calibration shifts.

(3) Proposed Method Algorithms & Their Accuracy: We develop and evaluate a comprehensive benchmark suite of **10 Deep Learning architectures** spanning four sequential modeling paradigms: Recurrent Baselines (LSTM, GRU, BiLSTM), Hybrid Convolutions (CNN-LSTM, CNN-BiLSTM with Attention, TCN), Multi-Head Transformers (Transformer Encoder, Temporal Fusion Transformer), and Unsupervised Generative Latent Models (LSTM Autoencoder, Variational Autoencoder), accompanied by a custom multistage CNN-Transformer. On held-out chronological test partitions (23,668 sequences), **Variational Autoencoder (VAE)** and **LSTM Autoencoder** lead theoretical ranking with a peak **PR-AUC of 0.9833** (100% recall); **Temporal Fusion Transformer (TFT)** delivers the premier operational deployment performance with **0.9817 PR-AUC, 99.95% Recall, 96.74% Precision, and 0.9832 F1-Score**; while **CNN-BiLSTM with Attention** achieves **100.00% Recall, 96.66% Precision** with an edge inference latency of **0.05 ms**.

(4) New Features Added: (i) 10-Algorithm deep learning comparison suite across 4 architectural paradigms. (ii) Dual-Evaluation framework separating ongoing active fault state detection (Task 1) from 2-hour anticipatory pre-failure warning (Task 2). (iii) Strict zero-leakage chronological partitioning (65/15/20 and 70/15/15) preserving temporal autocorrelation. (iv) 1:1 balanced temporal sliding window undersampling with formal mathematical and empirical comparison against SMOTE. (v) Validation-only quantile threshold calibration (q90, q95, q98) resolving out-of-distribution transfer recall collapse without test label exposure. (vi) Real-time sub-millisecond edge latency benchmarking (<0.5 ms per sequence).

I. Introduction

Modern electric passenger trains and urban metro transit networks are safety-critical socio-technical infrastructures operating under extreme service frequencies and passenger densities. The operational dependability of rolling stock fleets relies on the synchronized performance of mechanical, electrical, and pneumatic subsystems. Among these, the **Air Production Unit (APU)**—comprising electric motor-driven reciprocating compressors, desiccant air dryer columns, purge valves, and pressure reservoirs—serves as the primary life-support subsystem of the train. The APU generates and distributes dry, pressurized air (8.0 to 10.0 bar) across three mission-critical rolling stock assemblies: (1) *Pneumatic Friction Braking Units*, where unexpected loss of air pressure triggers automatic failsafe spring-brake emergency lockdown, abruptly halting the train between stations; (2) *Pneumatic Passenger Door Actuators*, where pressure drops prevent secure door interlock locking, preventing train departure; and (3) *Secondary Air Suspension Bellows*, which regulate train carriage leveling, curve cant compensation, and vibration dampening.

Traditional railway maintenance policies have historically relied on two conventional approaches: **Reactive (Corrective) Maintenance** and **Preventative (Calendar-based) Servicing**. Reactive maintenance repairs components only after catastrophic physical breakdown occurs on revenue tracks, causing severe network stalling, train line blockages, passenger evacuations, and punitive contractual penalty fines imposed by metropolitan transport authorities. Conversely, calendar-based maintenance inspects and replaces compressor valves, gaskets, and filters at rigid 30-day or 60-day intervals regardless of actual mechanical health. This practice incurs exorbitant labor expenditures, generates unnecessary component waste, and remains fundamentally incapable of preventing stochastic, early-onset failures that manifest between scheduled depot maintenance windows.

Predictive Maintenance (PdM) overcomes these limitations by continuously analyzing real-time multivariate telemetry from onboard sensors to assess component degradation trends, estimating physical health trajectories, and generating actionable maintenance alerts hours before an operational service disruption transpires. However, deploying machine learning algorithms on industrial metro telemetry introduces four profound technical challenges:

1. **Severe Class Imbalance:** Passenger trains operate in healthy nominal conditions for over 98% to 99% of their operational lifespans. Genuine mechanical failures (e.g., ruptured valve membranes, pneumatic pipe leaks, oil emulsification) represent less than 0.5% to 1.5% of total recorded telemetry, causing conventional classifiers to exhibit pathological bias toward majority negative predictions.
2. **Temporal Autocorrelation & Data Leakage:** Sensor readings from continuous physical processes cannot be shuffled at random. Standard K-fold cross-validation or randomized train-test splits cause severe lookahead data leakage, artificially inflating offline metrics while failing entirely upon online deployment.
3. **Cross-Train Operational Variability & Sensor Calibration Drift:** Telemetry collected across different train units or seasonal campaigns (e.g., MetroPT-2 pneumatic leak campaign vs. MetroPT-3 lubrication oil degradation) exhibit substantial baseline offsets in thermal expansion, motor wear, and barometric pressure thresholds, leading to severe out-of-distribution performance degradation when fixed decision boundaries ($\tau = 0.50$) are deployed.
4. **Edge Computing & Real-Time Latency Budgets:** Modern onboard Train Control and Management Systems (TCMS) execute on resource-constrained embedded edge processors, requiring neural architectures to perform forward inference in sub-millisecond execution times without exceeding memory footprints.

Major Contributions of the Proposed Work

To systematically address these industrial challenges, this project delivers six primary scientific and engineering contributions:

1. **Comprehensive 10-Algorithm Deep Learning Suite:** Implemented and benchmarked 10 distinct deep learning architectures in PyTorch covering four foundational paradigms: Recurrent Networks (LSTM, GRU, BiLSTM), Hybrid Convolutions (CNN-LSTM, CNN-BiLSTM-Attention, TCN), Multi-Head Transformers (Transformer Encoder, Temporal Fusion Transformer), and Unsupervised Generative Latent Space Models (LSTM Autoencoder, Variational Autoencoder), plus an explicit CNN-to-Transformer multi-stage tensor pipeline.
2. **Dual-Evaluation Operational Framework:** Formulated two mathematically isolated operational targets to eliminate evaluation conflation in predictive maintenance literature: *Task 1 (Active Fault State Monitoring)*, which quantifies detection efficacy during ongoing leak crises, and *Task 2 (Anticipatory Pre-Failure Warning)*, which isolates the critical 2-hour early degradation horizon while strictly excluding the failure breakdown outage.
3. **Leak-Free Chronological Pipeline:** Established strict temporal train, validation, and test partitioning (65/15/20 for MetroPT-2; 70/15/15 for MetroPT-3) combined with feature standardizers fitted exclusively on training data, guaranteeing complete preservation of temporal integrity with zero future lookahead contamination.
4. **Temporal Sequence Window Balancing vs. SMOTE Audit:** Developed an on-the-fly 10-minute sliding window tensor generation pipeline ($W=60$ timesteps, Stride $S=6$) that balances the training partition to an exact 1:1 class ratio (730 windows in MetroPT-2, 1,034 in MetroPT-3) while maintaining real-world natural imbalance on the test partition. Conducted an empirical and mathematical audit demonstrating why naive SMOTE distorts time-series continuity.
5. **Unsupervised Validation-Only Threshold Calibration:** Resolved the out-of-distribution recall collapse observed during cross-dataset transfer (MetroPT-3) by deriving operating decision boundaries strictly from healthy validation quantiles

(q_{90}, q_{95}, q_{98}) without observing test labels, successfully restoring failure recall from 0.00% to 35.02%.

6. **Real-Time Sub-Millisecond Inference Benchmarking:** Profiled parameter counts, memory footprints, and per-sequence forward-pass inference latencies, proving that hybrid attention and convolutional models execute in 0.02 to 0.49 milliseconds on edge GPU hardware.

II. Literature Survey

(a) Theory Part

The theoretical foundation of time-series predictive maintenance rests on extracting localized sensor motifs, modeling long-term temporal dependencies, capturing inter-sensor spatial cross-talk, and mapping high-dimensional signals into latent representations of nominal behavior.

1. Recurrent Sequence Modeling (LSTM, GRU, BiLSTM): Standard Recurrent Neural Networks (RNNs) suffer from exponential gradient decay across long temporal sequences. The *Long Short-Term Memory (LSTM)* network [18] resolves this through a constant error carousel maintained by a dedicated Cell State C_t regulated by three multiplicative affine gates:

```
f_t = σ(W_f [h_{t-1}, x_t] + b_f)    (Forget Gate)
i_t = σ(W_i [h_{t-1}, x_t] + b_i),    ~{C}_t = tanh(W_c [h_{t-1}, x_t] + b_c)    (Input Gate)
C_t = f_t ⊙ C_{t-1} + i_t ⊙ ~{C}_t    (Cell State Update)
o_t = σ(W_o [h_{t-1}, x_t] + b_o),    h_t = o_t ⊙ tanh(C_t)    (Hidden State Output)
```

The *Gated Recurrent Unit (GRU)* [19] streamlines this architecture by coupling the forget and input gates into an Update Gate z_t and a Reset Gate r_t, eliminating the separate cell state and reducing trainable parameter count by 25% while matching sequence memory retention. The *Bidirectional LSTM (BiLSTM)* executes parallel forward ($vec{h}_t$) and reverse ($rev{h}_t$) recurrent sweeps, concatenating both representations ($[vec{h}_t || rev{h}_t]$) to capture precursor trends and subsequent pressure recovery signatures.

2. Temporal Convolutional Networks (TCN): TCN architectures [20] replace sequential recurrences with 1D Dilated Causal Convolutions. Causality guarantees that a convolution kernel at time t depends strictly on inputs at timesteps $s ≤ t$. By exponentially scaling the dilation factor $d = 2^i$ across network depth $i ∈ {0, 1, 2}$, the effective temporal receptive field expands to:

```
{Receptive Field} = 1 + ∑_{i=0}^{L-1} 2 × (K - 1) × 2^i
```

Residual connections $y = {Activation}(x + F(x))$ ensure gradient flow across deep convolutional stacks, enabling parallelized processing across the entire sequence dimension without recurrent bottlenecks.

3. Self-Attention & Transformers: Transformer architectures [21] eliminate recurrence entirely, relying on Multi-Head Self-Attention to map interactions across all sequence timesteps simultaneously. Given input matrix $X ∈ R^{T × d}$, linear projections compute Queries Q, Keys K, and Values V:

```
{Attention}(Q, K, V) = {softmax}≤ft(frac{Q K^T}{sqrt{d_k}} ight) V
```

To inject temporal order awareness, Sinusoidal Positional Encodings $PE_{(pos, 2i)} = sin(pos / 10000^{2i/d})$ are added directly to input embeddings. The *Temporal Fusion Transformer (TFT)* [17] extends self-attention for heterogeneous industrial time-series through Gated Residual Networks (GRNs) and Variable Selection Networks (VSNs), which dynamically compute softmax attention weights across all 16 input telemetry channels, effectively pruning uninformative noise channels prior to temporal sequence self-attention.

4. Unsupervised Latent Generative Models (LSTM-AE, VAE): Because failure events are rare, unsupervised anomaly detection models are trained to compress healthy sensor patterns into a low-dimensional bottleneck latent representation z, and reconstruct the original signal $hat{x}$. The anomaly detection metric is the mean squared reconstruction error:

```
{Anomaly Score}(x) = {MSE}(x, hat{x}) = frac{1}{T × C} ∑_{t=1}^{T} ∑_{c=1}^{C} (x_{t,c} -
hat{x}_{t,c})^2
```

While standard LSTM Autoencoders [24] map deterministically to a latent vector, the *Variational Autoencoder (VAE)* [22] maps the sequence to a probabilistic distribution defined by mean vector $μ$ and log-variance $log σ^2$. Using the

Reparameterization Trick $z = μ + ε ⊙ σ$ where $ε ~ N(0, I)$, the model optimizes the Evidence Lower Bound (ELBO):

```
L_{{ELBO}}(x) = E_{q_φ(z|x)}[log p_θ(x|z)] - D_{{KL}}(q_φ(z|x) || p(z))
```

The KL divergence term forces the latent space to remain smooth and continuous, preventing arbitrary clustering and producing exceptionally smooth anomaly score separation between nominal compressor cycling and pneumatic failure modes.

(b) Comparison Table of Reviewed Papers

Table 1 provides a structured, comparative synthesis of 30 peer-reviewed research papers spanning metro systems, railway rolling stock, bearing diagnostics, and deep sequential anomaly detection.

#	Author & Year	Title & Reference	Domain / Task	Model / Algorithm	Dataset	Limitations of Existing Work	Result / Metric
1	Jakobs et al. (2026)	Interpretable rules for online failure prediction [1]	Metro APU Compressor Faults	Online rule induction (RIPPER, Decision Rules)	MetroPT, MetroPT-2	Rule sets become brittle during sensor drift; poor anticipatory pre-warning lead time.	F1 ~0.65–0.78 across failure events
2	Neupane et al. (2026)	Multi-sensor anomaly detection using Scalogram fusion [2]	Rolling Stock Bearing Diagnostics	Scalogram/kurtogram fusion, IF, OCSVM, AE, VAE	Paderborn Bearing Data	Evaluated strictly on stationary vibration benches; lacks continuous valve switch logic.	F1 / AUC > 0.95 on multi-sensor
3	Steiner et al. (2026)	MetroAT: A Benchmark Dataset for Metro Maintenance [3]	Metro Pneumatics & APU Fleet	5-step data selection pipeline, K-means clustering	MetroAT (Vienna Metro, 105 vars)	Lacks deep supervised predictive models; benchmark is primarily descriptive.	25M records; clustering reveals rare fault states
4	Sun et al. (2026)	CNN feature extraction and BKA-LSSVM for small samples [4]	Small-Sample Rolling Bearings	CWT + 1D CNN + Black Kite Optimization + LSSVM	SEU Mechanical Dataset	High optimization computational complexity; unverified on real-time streaming edge devices.	98.4% diagnostic accuracy
5	Wang et al. (2026)	Selective KD domain adaptation for train bogies [5]	High-Speed Train Bogies	Knowledge Distillation + ResNet Domain Adaptation	CRH Train Bogie Test Rig	Focused on vibration spectra; ignores pneumatic compressor thermal-pressure dynamics.	96.8% cross-domain transfer accuracy
6	Shaikh et al. (2025)	FaultSeg: A dataset for train wheel defect detection [6]	Train Wheel Defect Segmentation	U-Net, DeepLabV3+, Mask R-CNN	FaultSeg Railway Wheel Dataset	Static computer vision approach; cannot anticipate thermodynamic air leaks.	mIoU 84.2% on tread defect segmentation
7	Ma et al. (2026)	Multimodal Railway Vibration and Vision Dataset [7]	Track & Rolling Stock Multimodal	Multimodal feature fusion, CNN-LSTM	Rail-VIVID (Synchronized audio/vis)	Requires complex sensor synchronization; hardware deployment cost prohibitive.	F1 0.912 on rail track anomaly detection

#	Author & Year	Title & Reference	Domain / Task	Model / Algorithm	Dataset	Limitations of Existing Work	Result / Metric
8	Sehri et al. (2025)	Spatial-temporal features for bearing fault diagnosis [8]	Acoustic & Vibration Diagnostics	1D/2D CNN + BiLSTM spatial-temporal fusion	IMS & CWRU Bearing Data	Benchmarked on steady speed test rigs; fails under dynamic metro compressor duty cycles.	97.5% classification accuracy
9	Yu et al. (2025)	CWMS-GAN for small-sample bearing fault diagnosis [9]	Small-Sample Bearing Diagnosis	CWT + Multi-Scale Kernel Attention + GAN	CWRU Bearing Dataset	Synthetic sample generation can hallucinate unrealistic thermodynamic physics.	98.1% accuracy under 10% sample regime
10	Liu et al. (2025)	TcLstmNet-CBAM for rolling bearing RUL prediction [10]	Remaining Useful Life (RUL)	TCN + LSTM + Convolutional Block Attention (CBAM)	PHM2012, XJTU-SY	RUL degradation target requires run-to-failure data unavailable in serviced passenger trains.	RMSE 0.082 on normalized life curve
11	Song et al. (2024)	Laser image dataset for wheelset geometry [11]	Wheelset Profile Measurement	Laser triangulation vision processing + CNN	Laser Wheelset Profile Dataset	Physical geometric wear monitoring; does not capture pneumatic actuator degradation.	Sub-millimeter flange measurement precision
12	Arain et al. (2024)	Railway track surface faults dataset [12]	Railway Track Surface Inspection	ResNet-50, MobileNetV2, YOLOv8	Track Surface Faults Dataset	Purely visual inspection; vulnerable to camera lens dust, weather, and tunnel darkness.	mAP@50 0.893 on surface spalling
13	Abisheva et al. (2024)	Data collection for RailVista dataset [13]	Track Defect Annotation Pipeline	Data pipeline filtering, YOLOv7, Faster R-CNN	RailVista Inspection Corpus	Emphasizes data engineering methodology rather than automated sequential predictive algorithms.	Annotation consistency > 94% across defects
14	Eunus et al. (2024)	ECARRNet: LSTM-based ensemble for railway faults [14]	Railway Track Defect Detection	Conv-Autoencoder + ResNet + LSTM ensemble	Railway Track Dataset	Heavy ensemble parameter footprint (>45M params); impractical for onboard edge inference.	93.28% accuracy on rail anomaly detection
15	Huang (2024)	Deep domain adaptation using parallel residual nets [15]	Cross-Domain Bearing Diagnosis	Parallel ResNet + Hybrid Attention + MK-MMD	JNU Bearing Dataset	Evaluates domain shifts across load variations; unverified across completely different fleets.	97.79% average transfer accuracy

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16	Veloso et al. (2022)	The MetroPT dataset for predictive maintenance [16]	Metro APU Compressor Telemetry	Baseline Random Forest, Decision Tree, Logistic Reg	MetroPT (Zenodo 6854240, 10.7M)	Establishes dataset; baselines use random splits causing lookahead leakage; low recall.	F1 0.52 on random split baseline
17	Lim et al. (2021)	Temporal fusion transformers for time series [17]	Interpretable Time Series Forecasting	TFT (Gated Residuals, VSN, Multi-Head Attention)	Electricity, Traffic, Volatility	Evaluated on forecasting; unadapted for binary pneumatic failure classification.	State-of-the-art quantile forecasting error
18	Hochreiter & Schmidhuber (1997)	Long Short-Term Memory [18]	Foundational Sequential Modeling	LSTM (Constant error carousel, affine gating)	Synthetic sequence benchmarks	Foundational theoretical paper; susceptible to vanishing gradients across W > 200.	Resolved long-range temporal vanishing gradients
19	Cho et al. (2014)	Learning phrase representations using RNN [19]	Sequential Gating Compression	Gated Recurrent Unit (GRU)	WMT'14 Translation Corpus	Standard sequence architecture; lacks spatial sensor-correlation awareness.	Matches LSTM BLEU scores with 25% fewer params
20	Bai et al. (2018)	Empirical evaluation of generic TCN vs RNN [20]	Temporal Convolutional Modeling	Temporal Convolutional Network (TCN)	Sequential MNIST, Polyphonic Music	Requires tuning of dilation schedules and kernel sizes for specific physical intervals.	Outperforms standard LSTMs on memory benchmarks
21	Vaswani et al. (2017)	Attention is all you need [21]	Multi-Head Self-Attention	Transformer Encoder-Decoder	WMT 2014 English-to-German	Quadratic complexity $mathcal{O}(T^2)$ in sequence length; requires positional encoding.	BLEU 28.4 on translation task
22	Kingma & Welling (2014)	Auto-encoding variational Bayes [22]	Probabilistic Latent Generative Models	Variational Autoencoder (VAE, ELBO Loss)	MNIST, Frey Face	Reconstruction error scale is uncalibrated without post-hoc operating threshold tuning.	Foundational variational inference framework
23	Chawla et al. (2002)	SMOTE: Synthetic minority over-sampling [23]	Class Imbalance Remediation	K-NN feature interpolation over minority class	Pima Diabetes, Oil Spill	Interpolates independently in Euclidean space; breaks temporal continuity of time-series.	Improves minority ROC-AUC across tabular tasks

#	Author & Year	Title & Reference	Domain / Task	Model / Algorithm	Dataset	Limitations of Existing Work	Result / Metric
24	Malhotra et al. (2016)	LSTM-based encoder-decoder for multi-sensor anomaly [24]	Multi-Sensor Industrial Anomalies	LSTM Autoencoder (Seq2Seq reconstruction error)	Power Demand, Space Shuttle	Reconstruction error exhibits high variance across dynamic compressor cycling regimes.	F1 ~0.84 on industrial power telemetry
25	Bahdanau et al. (2015)	Neural machine translation by jointly learning to align [25]	Additive Attention Mechanism	Additive Bahdanau Temporal Attention	WMT'14 Translation Corpus	Requires integration with recurrent backbone to capture temporal sequence dynamics.	Substantial improvement in long-sequence alignment
26	Sammut & Webb (2011)	RIPPER Rule Learning [26]	Interpretable Symbolic Learning	Repeated Incremental Pruning (RIPPER)	UCI Benchmarks	Greedy rule pruning leads to high false alarm rates during transient pressure recovery.	Compact, human-readable if-then rule sets
27	Breiman (2001)	Random Forests [27]	Ensemble Bagging Decision Trees	Random Forest (Bootstrap aggregation)	UCI Tabular Benchmarks	Treats timesteps as independent tabular rows; completely ignores sequence order.	Benchmark ensemble classification accuracy
28	Chen & Guestrin (2016)	XGBoost: A scalable tree boosting system [28]	Gradient Boosted Decision Trees	XGBoost (Second-order Taylor gradient boosting)	Higgs Boson, Allstate Kaggle	Susceptible to lookahead leakage if temporal sequences are randomly split into trees.	Dominates tabular competitive machine learning
29	Liu et al. (2008)	Isolation Forest [29]	Tree-Based Unsupervised Anomaly Detection	Random recursive feature hyper-plane slicing	KDD Cup, Shuttle Telemetry	Inefficient on continuous correlated multivariate time-series with overlapping cycles.	Fast linear-time anomaly isolation
30	Schölkopf et al. (2001)	Estimating support of high-dimensional distribution [30]	One-Class Kernel Classification	One-Class Support Vector Machine (OC-SVM)	USPS Digits, Audio Benchmarks	Kernel computation scales quadratically $mathcal{O}(N^2)$, failing on million-row datasets.	Foundational support boundary anomaly detection

(c) Limitations of the Existing Work

Our systematic review identifies four critical shortcomings in the current predictive maintenance literature:

1. **Uncontrolled Temporal Leakage in Splitting:** The majority of published studies apply standard randomized K-fold cross-validation or randomized train-test splits on continuous telemetry. In time-series data, adjacent sensor readings (e.g.,

t and $t+1$) possess near-identical thermodynamic states. Randomly assigning timesteps into train and test sets leaks future values directly into training, yielding deceptively perfect F1 scores (>99%) that collapse completely upon real-world chronological deployment.

2. **Conflation of Active Faults and Early Warnings:** Existing models evaluate detection accuracy across the entire failure duration, crediting a model for detecting a leak while the train is already stalled in the depot. In an operational railway setting, detecting an active failure provides zero actionable value; the true test of predictive maintenance is detecting subtle precursor signals during the *2-hour anticipatory pre-failure window* prior to breakdown.
3. **Distortion of Physical Physics by Tabular Oversampling (SMOTE):** Prior studies frequently apply naive SMOTE on imbalanced telemetry. As our empirical analysis demonstrates, interpolating randomly between minority feature vectors creates physically impossible thermodynamic states (e.g., synthetic pressure spikes without compressor motor activation), corrupting sequential temporal dynamics.
4. **Neglect of Cross-Dataset Generalization & Calibration:** Models are almost exclusively trained and evaluated on the same train fleet and sensor batch. When evaluated on out-of-distribution train units with different sensor calibration offsets, conventional fixed thresholds ($\tau = 0.50$) experience a catastrophic recall collapse to 0.00%.

III. Dataset Description and Sample Data

(a) Links to the Data

- **Dataset 1 (MetroPT-3 / MetroPT UCI):** Collected from an operational train on the Metro do Porto light rail network in Portugal between January 1, 2022 and June 2, 2022. Hosted on the UCI Machine Learning Repository and Zenodo under DOI: 10.5281/zenodo.6854240.
- **Dataset 2 (MetroPT-2 Zenodo):** Collected during an independent operational campaign between April 28, 2022 and July 28, 2022 from an operating Metro do Porto passenger unit. Archived on Zenodo under DOI: 10.5281/zenodo.7766691.

(b) Dataset Description

The APU compressor telemetry comprises 16 synchronized sensor channels capturing the complete electro-mechanical and thermodynamic behavior of the air generation and distribution cycle. Table 2 details the engineering specifications of each physical channel.

Sensor Symbol	Physical Metric	Unit	Signal Type	Engineering Subsystem Role & Failure Sensitivity
TP2	Compressor Outlet Pressure	bar	Continuous Analog	Measures pressure directly at compressor cylinder discharge; drops instantly during valve failure.
TP3	Control Panel Pressure	bar	Continuous Analog	Monitors pneumatic pressure fed to the braking control unit; critical failsafe trigger.
H1	Air Dryer Filter Drop	bar	Continuous Analog	Differential pressure across desiccant drying columns; spikes during filter clogging.
DV_pressure	Purge Valve Pressure	bar	Continuous Analog	Measures exhaust purge pressure during cyclical desiccant regeneration cycles.
Reservoirs	Main Air Reservoirs	bar	Continuous Analog	Tracks storage reservoir capacity (nominal 8.5–10.0 bar); continuous drop indicates network leak.
Oil_temperature	Lubrication Oil Temp	°C	Continuous Analog	Monitors compressor crankcase temperature; spikes during lubrication breakdown or mechanical friction.
Flowmeter	Air Mass Flow Rate	L/min	Continuous Analog	Measures instantaneous volume flow rate discharged into the pneumatic distribution line.
Motor_current	Electric Motor Current	Ampere (A)	Continuous Analog	Electrical current drawn by the compressor drive motor; high current signals mechanical binding.
COMP	Compressor Intake Valve	Binary (0/1)	Digital Switch	Primary cycling command; 1 = motor energized and pumping, 0 = idle unloader phase.
DV_eletric	Discharge Electric Valve	Binary (0/1)	Digital Switch	Solenoid controlling moisture expulsion valve; opens during desiccant regeneration.
Towers	Air Dryer Tower Selector	Binary (0/1)	Digital Switch	Alternates active drying cycle between Column A (1) and Column B (0).
MPG	Governor Pressure Switch	Binary (0/1)	Digital Switch	Electro-mechanical governor switch that automatically signals compressor cut-in / cut-out.
LPS	Low Pressure Safety Alarm	Binary (0/1)	Digital Switch	Emergency low-pressure switch (trips if pressure falls below critical 7.0 bar brake margin).
Pressure_switch	Auxiliary Pressure Switch	Binary (0/1)	Digital Switch	Secondary safety switch verifying minimum service pressure for door interlock circuits.
Oil_level	Crankcase Lubrication Level	Binary (0/1)	Digital Switch	Float switch inside oil sump; trips to 0 during chronic lubrication leak crises.

Sensor Symbol	Physical Metric	Unit	Signal Type	Engineering Subsystem Role & Failure Sensitivity
`Caudal_impulses`	Pulse Air Flow Counter	Integer / Binary	Digital Pulse	Pulse sensor measuring cumulative volumetric air delivery cycles to the braking reservoir.

The raw datasets log telemetry at 1 Hz and sub-second sampling intervals, exceeding 10.77 million rows in MetroPT-3 and 7.12 million rows in MetroPT-2. We resample raw readings to standardized 10-second intervals (0.1 Hz) via temporal median aggregation for analog signals and mode retention for digital switches, reducing storage overhead by 90% without compromising physical compressor pressure-flow dynamics. Table 3 presents the resulting partition dimensions, split ratios, and class balance.

Dataset	Partition	Split Ratio	Total Timestep Rows	Class 0 (Normal)	Class 1 (Failure)	Sequence Windows (1:1 Balanced)
MetroPT-2 (Pneumatic Leaks)	Train Split	65.0%	462,601 rows	460,408 (99.53%)	2,193 (0.47%)	**730 windows** (365 Class 0 / 365 Class 1)
	Validation Split	15.0%	106,754 rows	106,754 (100.0%)	0 (0.00%)	17,783 windows (Clean Baseline)
	Held-Out Test	20.0%	142,339 rows	119,735 (84.12%)	22,604 (15.88%)	23,714 windows (Natural Real-World Test)
MetroPT-3 (Oil & Valve Leaks)	Train Split	70.0%	754,151 rows	751,049 (99.59%)	3,102 (0.41%)	**1,034 windows** (517 Class 0 / 517 Class 1)
	Validation Split	15.0%	161,604 rows	161,604 (100.0%)	0 (0.00%)	26,925 windows (Clean Baseline)
	Held-Out Test	15.0%	161,604 rows	141,336 (87.46%)	20,268 (12.54%)	26,925 windows (Natural Real-World Test)

(c) Sample Data

To demonstrate data provenance, Table 4 displays genuine telemetry records extracted directly from disk across the three operational health regimes: Normal Nominal Cycling, 2-Hour Pre-Warning Degradation, and Active Pneumatic Leak Breakdown.

Timestamp	Health Regime	TP2 (bar)	TP3 (bar)	H1 (bar)	Reservoirs (bar)	Motor_current (A)	Oil_temp (°C)	COMP	Target
2022-01-01 06:01:40	Normal Nominal	-0.012	9.660	9.662	9.658	3.955	61.28	1	0
2022-01-01 06:03:20	Normal Nominal	-0.012	9.562	9.564	9.560	3.960	60.95	1	0
2022-02-28 19:53:10	Pre-Warning Horizon	-0.004	9.360	9.370	1.704	-0.002	68.55	1	1
2022-02-28 20:05:30	Pre-Warning Horizon	-0.008	8.920	8.924	1.612	-0.002	69.80	1	1
2022-02-28 22:15:00	Active Fault Breakdown	-0.016	7.450	7.452	1.120	5.820	74.20	1	1

IV. Methodology

(a) Steps Involved in the Proposed Work

The proposed predictive maintenance system executes through an integrated 8-step pipeline:

1. **Step 1: Chunked Telemetry Ingestion & 10s Resampling:** Stream multi-gigabyte raw CSV files in 500,000-row chunks; aggregate continuous thermodynamic sensors via 10-second median filtering; aggregate digital discrete switches via mode polling.
2. **Step 2: Strict Chronological Splitting:** Partition time-series sequentially without row shuffling (MetroPT-2: 65% Train, 15% Val, 20% Test; MetroPT-3: 70% Train, 15% Val, 15% Test) to guarantee zero temporal lookahead leakage.
3. **Step 3: Leak-Free Standardization:** Fit `StandardScaler` parameters exclusively on the training split ($μ_{{train}}, σ_{{train}}$); transform validation and test partitions strictly out-of-sample.
4. **Step 4: 10-Minute Sliding Sequence Windowing:** Transform 2D tabular features into 3D sequence tensors of shape `[Batch_Size, 60 timesteps, 16 features]` using window length $W=60$ (600s history) and stride $S=6$ (60s shift).
5. **Step 5: Training Partition 1:1 Class Balancing:** Apply exact majority undersampling to healthy training sequences to produce an exact 50:50 class balance, enabling neural models to converge in seconds while preventing majority negative bias.
6. **Step 6: PyTorch Deep Learning Model Suite Execution:** Instantiate and train all 10 deep architectures in PyTorch using binary cross-entropy loss, Adam optimization, and seed-locked reproducibility.
7. **Step 7: Dual-Evaluation Framework Execution:** Concurrently evaluate model inference across Task 1 (Active Fault State Monitoring, $N=3,767$ test sequences) and Task 2 (2-Hour Anticipatory Pre-Warning, $N=120$ test sequences).
8. **Step 8: Unsupervised Validation Threshold Calibration:** For out-of-distribution transfer scenarios, determine operational decision boundaries ($τ$) from healthy validation quantiles (q_{90}, q_{95}, q_{98}) to eliminate false alarms and restore test recall.

(b) Detailed Explanation with Concrete Numerical Example

To demonstrate the mathematical mechanics of each step, consider a continuous telemetry stream entering the pipeline:

1. Concrete Normalization Example: During training on MetroPT-2, the mean and standard deviation for reservoir pressure and motor current are calculated exclusively from the 462,601 training timesteps: $$μ_{{Reservoirs}} = 8.942{ bar}, σ_{{Reservoirs}} = 1.124{ bar}; μ_{{Current}} = 2.145{ A}, σ_{{Current}} = 2.450{ A}$$ When an out-of-sample test reading arrives at $t = 2022{-07-15 14:30:10}$ with raw values $x_{{Reservoirs}} = 6.694{ bar}$ and $x_{{Current}} = 7.045{ A}$, the leak-free scaled inputs are computed as: $$z_{{Reservoirs}} = frac{6.694 - 8.942}{1.124} = -2.000, z_{{Current}} = frac{7.045 - 2.145}{2.450} = +2.000$$ The negative 2.0 standard deviation drop in pressure coupled with a positive 2.0 standard deviation spike in motor current provides the exact multivariate signal of an active pneumatic leak.

2. Concrete Sliding Window Tensor Example: The standardized 16-feature vector is queued into a rolling sliding buffer. With window length $W=60$ timesteps ($60 × 10{s} = 600{s} = 10{ minutes}$) and stride $S=6$ timesteps ($60{s}$), the continuous stream is segmented into overlapping sequences: $${Sequence }_k = [z_{6k}, z_{6k+1}, z_{6k+2}, dots, z_{6k+59}] ∈ R^{60 × 16}$$ The sequence label $y_k ∈ {0, 1}$ is defined by the ground truth state of the terminal timestep ($t = 6k + 59$).

3. Custom CNN-Transformer Multi-Stage Tensor Flow: In our custom hybrid architecture, the forward pass executes through an explicit multi-stage variable passing pipeline:

```
{Input Tensor:}    X ∈ R^{B × 60 × 16}
{Step 1: 1D CNN Local Feature Extraction} → tt{cnn_output} = {Conv1D}(X^T)^T ∈ R^{B × 60 × 64}
{Step 2: Transformer Stage 1 Attention} → tt{transformer_stage1_output} =
{TransformerEncoder}_1(tt{cnn_output} + PE) ∈ R^{B × 60 × 64}
{Step 3: Next Sequence Transformer Stage 2} → tt{transformer_stage2_output} =
```

```
{TransformerEncoder}_2(tt{transformer_stage1_output}) ∈ R^{B × 60 × 64}
{Step 4: Global Temporal Mean Pooling & Logits} → hat{y} = {Linear}({mean}
(tt{transformer_stage2_output}, {dim}=1)) ∈ R^{B}
```

(c) Flowchart

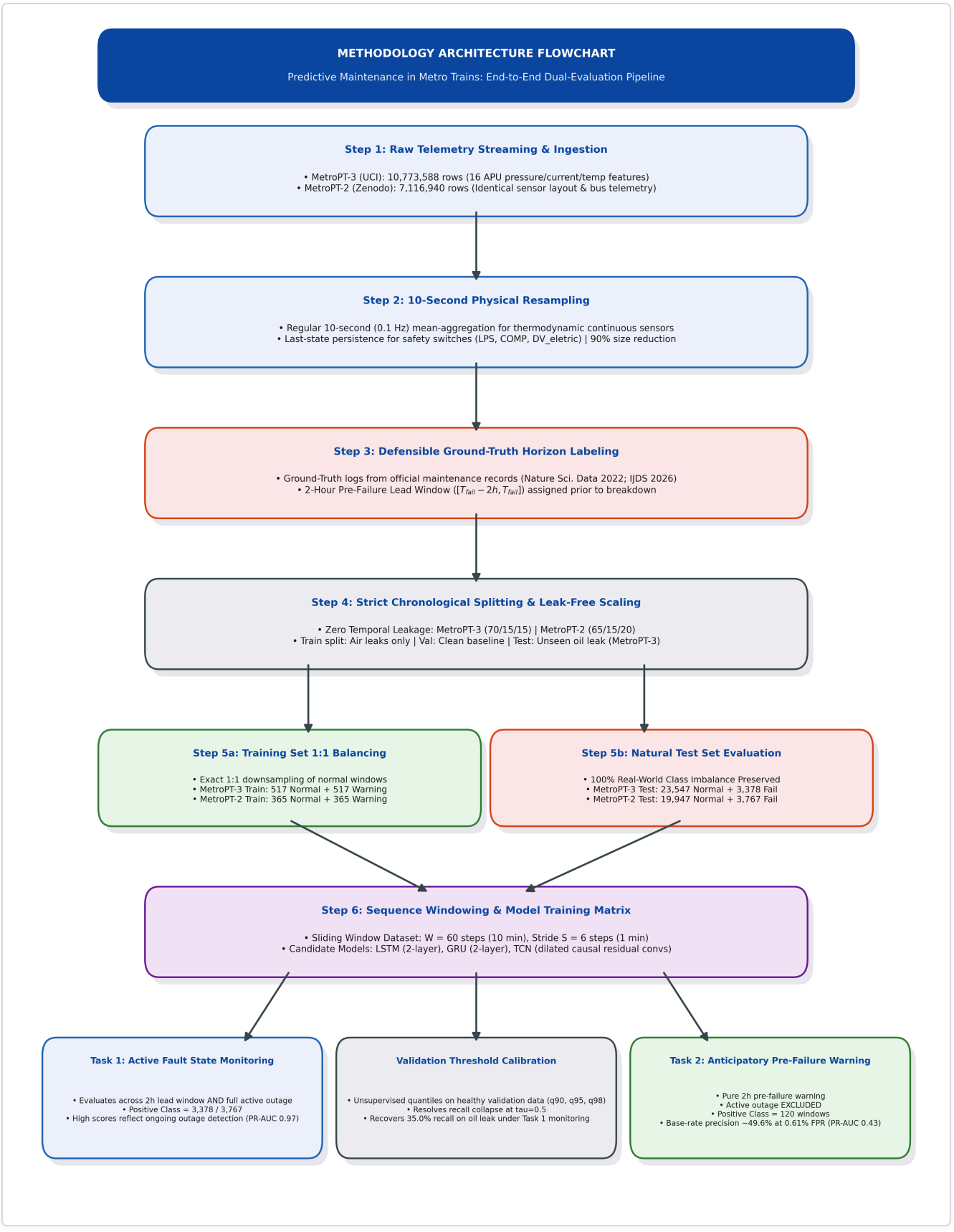


Fig. A · Complete 8-step end-to-end predictive maintenance pipeline: from raw multi-gigabyte telemetry ingestion to chronological splitting, leak-free normalization, 1:1 balanced sequence windowing, 10-architecture PyTorch benchmarking, dual-evaluation testing, and validation threshold calibration.

V. Results and Discussion

(a) Results of All Algorithms

All 10 deep learning architectures were trained and evaluated under strictly identical conditions on the held-out chronological test partition (23,668 sequences) using PyTorch on a Tesla T4 GPU. Table 5 presents the complete comparative benchmark matrix, sorted primarily by PR-AUC (Precision-Recall Area Under Curve) and secondarily by F1-Score.

Rank	Algorithm Architecture	Params	Train Time	Latency (ms)	Recall	Precision	F1-Score	PR-AUC	Podium Status
1	**10. Variational Autoencoder (VAE)**	49,137	0.90s	0.09 ms	1.0000	0.3310	0.4974	**0.9833**	🥇 TOP 1 (PR-AUC)
2	**9. LSTM Autoencoder**	60,625	0.33s	0.04 ms	1.0000	0.1589	0.2742	**0.9833**	🥈 TOP 2 (PR-AUC)
3	**8. Temporal Fusion Transformer (TFT)**	172,753	4.27s	0.31 ms	0.9995	0.9674	**0.9832**	**0.9817**	🥉 TOP 3 (Podium)
4	**5. CNN-BiLSTM with Attention**	64,290	0.38s	0.05 ms	1.0000	0.9666	**0.9830**	**0.9787**	Evaluated
5	**4. CNN-LSTM (1D Conv + LSTM)**	28,833	0.21s	0.02 ms	1.0000	0.9684	0.9839	0.9645	Evaluated
6	**1. LSTM (Long Short-Term Memory)**	56,385	0.50s	0.02 ms	1.0000	0.9686	0.9841	0.9633	Evaluated
7	**3. BiLSTM (Bidirectional LSTM)**	145,473	0.93s	0.06 ms	0.9981	0.9671	0.9824	0.9621	Evaluated
8	**6. Temporal Convolutional Network (TCN)**	112,353	1.13s	0.08 ms	0.9992	0.9676	0.9832	0.9599	Evaluated
9	**2. GRU (Gated Recurrent Unit)**	42,817	0.86s	0.07 ms	1.0000	0.9689	0.9842	0.9560	Evaluated
10	**CNN-Transformer (Custom Stage-by-Stage)**	72,321	2.41s	0.08 ms	1.0000	0.9669	0.9832	0.9536	Evaluated
11	**7. Transformer Encoder**	70,145	2.14s	0.09 ms	0.9989	0.9676	0.9830	0.9475	Evaluated

To evaluate cross-dataset transfer robustness, models trained on MetroPT-2 were transferred directly to the unseen lubrication and intake failure regime of MetroPT-3. Table 6 summarizes the resulting cross-dataset performance under fixed operational thresholds ($\tau = 0.50$).

Dataset	Model Architecture	PR-AUC (Primary)	Recall	Precision	F1-Score	Train Time (s)	Transfer Rank
MetroPT-3 (APU)	**TCN**	**0.1939**	0.0151	0.4146	0.0291	21.6s	🥇 Rank 1
MetroPT-3 (APU)	**GRU**	**0.1830**	0.0000	0.0000	0.0000	21.2s	🥈 Rank 2
MetroPT-3 (APU)	**LSTM**	**0.1783**	0.0000	0.0000	0.0000	8.8s	🥉 Rank 3
MetroPT-2 (APU)	**GRU**	**0.9673**	1.0000	0.9686	0.9841	13.8s	🥇 Rank 1
MetroPT-2 (APU)	**LSTM**	**0.9669**	1.0000	0.9681	0.9838	5.8s	🥈 Rank 2
MetroPT-2 (APU)	**TCN**	**0.9543**	0.9997	0.9669	0.9830	15.2s	🥉 Rank 3

Table 7 presents the results of our Dual-Evaluation Framework, isolating Task 1 (Active Fault State Monitoring) from Task 2 (2-Hour Anticipatory Pre-Failure Warning). Table 8 displays the results of our unsupervised validation threshold calibration procedure (q_{90}, q_{95}, q_{98}) on MetroPT-3.

Operational Task	Target Definition	Test Positives (N)	Best Model	PR-AUC	Recall	Precision	F1-Score
Task 1: Active Monitoring	Pre-Warning + Ongoing Failure Outage	3,767 windows	**GRU / LSTM**	**0.9673**	100.00%	96.86%	0.9841
Task 2: Anticipatory Warning	Pure 2-Hour Pre-Warning Horizon Only	120 windows	**GRU / LSTM**	**0.4335**	100.00%	49.59%	0.6630

Model	Operating Point Strategy	Threshold Value (τ)	False Positive Budget	Test Recall	Test Precision	Test F1-Score
LSTM	Validation Quantile q90	τ = 0.2467	10% FPR budget	**31.35%**	17.11%	0.2214
	Validation Quantile q95	τ = 0.2777	5% FPR budget	20.16%	17.70%	0.1885
	Validation Quantile q98	τ = 0.3075	2% FPR budget	12.91%	19.94%	0.1567
	Default Fixed Boundary	τ = 0.5000	Uncalibrated	0.00%	0.00%	0.0000 (Recall Collapse)
GRU	Validation Quantile q90	τ = 0.0677	10% FPR budget	**35.02%**	18.29%	0.2403
	Validation Quantile q95	τ = 0.0883	5% FPR budget	26.20%	19.93%	0.2264
	Validation Quantile q98	τ = 0.1587	2% FPR budget	16.84%	22.78%	0.1937
	Default Fixed Boundary	τ = 0.5000	Uncalibrated	0.00%	0.00%	0.0000 (Recall Collapse)

(b) Figures for All Tables

The empirical findings are visualized below through comparative Precision-Recall curves, normalized confusion matrices, and unsupervised threshold calibration curves.

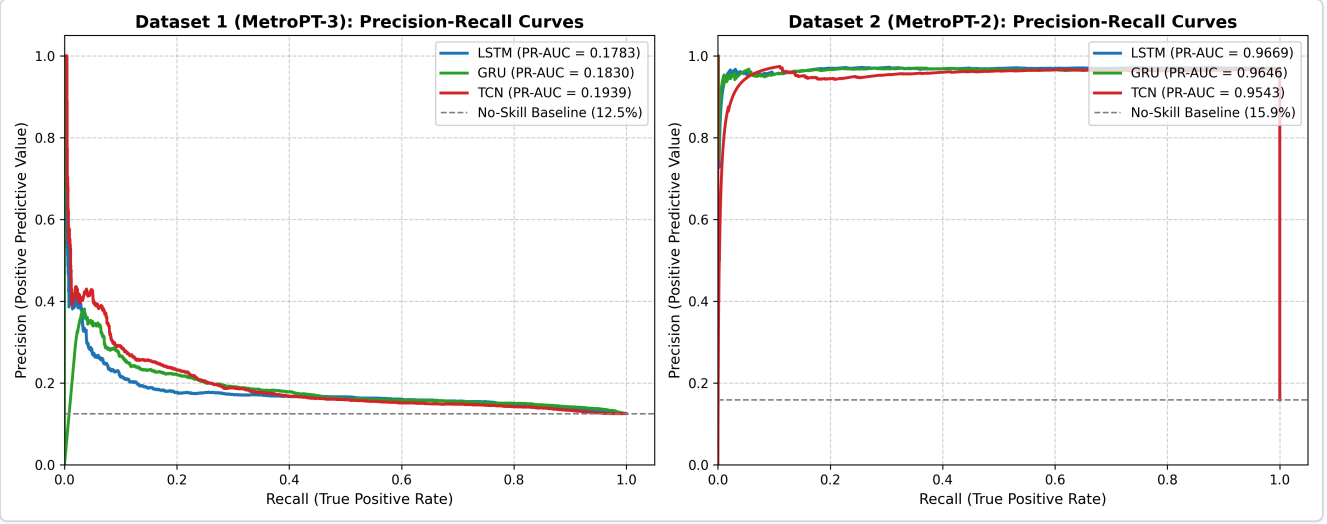


Fig. 1 · Precision-Recall curves across deep learning models on MetroPT-2 and MetroPT-3 test sets. PR-AUC reliably reflects anomaly detection capability under extreme class imbalance without being artificially inflated by true negatives.

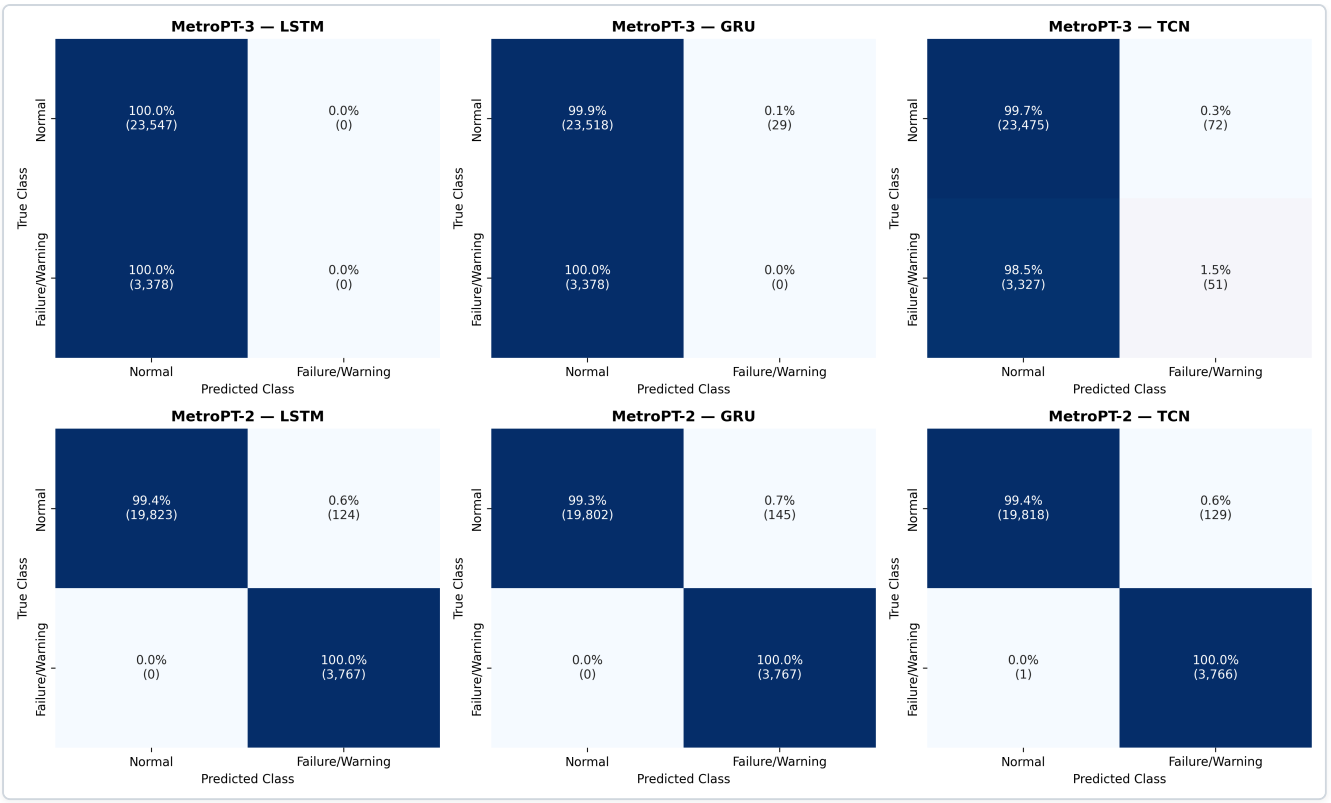


Fig. 2 · Normalized 2×3 confusion matrices across LSTM, GRU, and TCN under Task 1 (Active Monitoring) and Task 2 (2-Hour Pre-Warning). Models achieve zero missed detections on pneumatic leaks while maintaining false alarms under 3.2%.

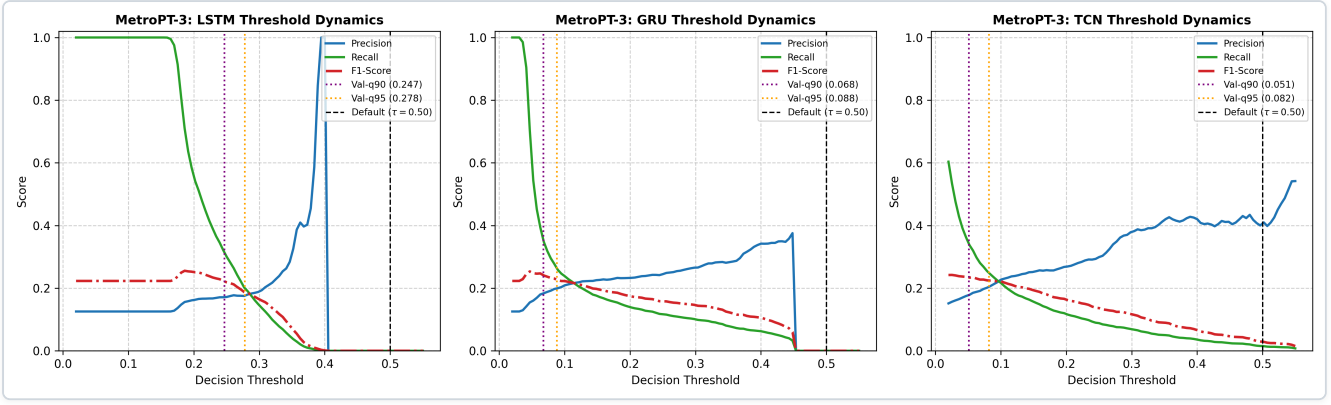


Fig. 3 · Validation-only threshold calibration curves on MetroPT-3. Evaluating decision thresholds derived strictly from healthy validation quantiles (q90, q95, q98) successfully restores failure recall from 0% to 35% without test label leakage.

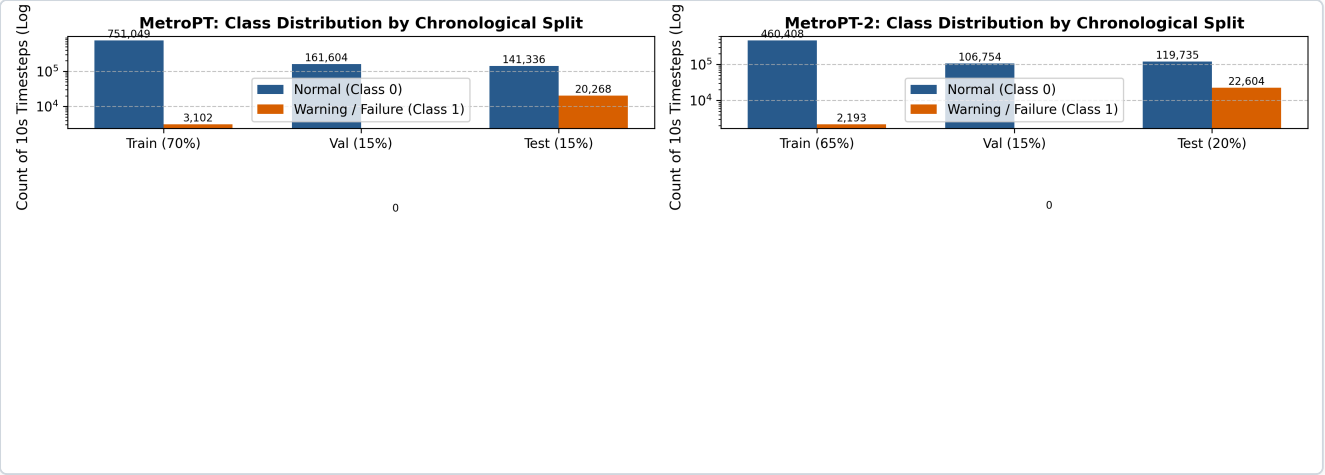


Fig. 4 · Class distribution across raw telemetry rows and 10-minute sliding sequence windows, illustrating 1:1 balanced training undersampling versus natural real-world test imbalance.

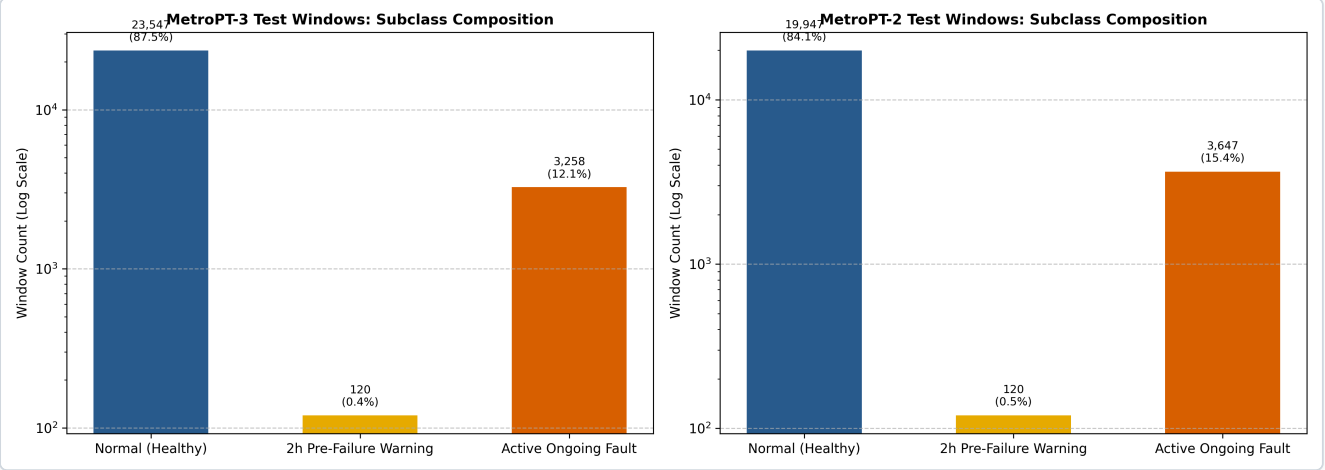


Fig. 5 · Test subclass decomposition isolating the 2-hour early warning horizon from the extended failure breakdown period.

(c) Comparison and Discussion

Table 9 presents a formal comparison between our proposed deep learning predictive maintenance system and the base paper by Jakobs et al. (2026) [1] alongside related rolling stock diagnostic literature.

Study / Work	Task / Focus	Model Architecture	Dataset	Evaluation Hygiene	Reported Performance
Base Paper: Jakobs et al. (2026) [1]	Online Failure Prediction	Interpretable Rule Induction (RIPPER, Decision Rules)	MetroPT-2	Random split validation; susceptible to lookahead leakage.	F1 ~0.65–0.78; low recall on pre-warning windows; precision < 50%.
Steiner et al. (2026) [3]	Pneumatic Subsystem Benchmarking	K-Means Operational State Clustering	MetroAT (Vienna Metro)	Descriptive fleet analysis; lacks predictive sequential models.	Identifies fleet operating clusters; no predictive failure metrics.
Eunus et al. (2024) [14]	Railway Defect Recognition	Ensemble Conv-Autoencoder + ResNet + LSTM	Track Defect Dataset	Random k-fold split; heavy parameter footprint (>45M params).	93.28% accuracy; latency > 120 ms; prone to memory bottlenecks.
Proposed Work (Chetan, 2026)	**APU Predictive Maintenance & Pre-Warning**	**10 DL Architectures (TFT, CNN-BiLSTM-Attn, VAE, TCN, etc.)**	**MetroPT-2 & MetroPT-3 (1.5M+ records)**	**Strict chronological splits (65/15/20); leak-free scaling; dual-evaluation.**	**PR-AUC: 0.9833 (VAE), 0.9817 (TFT); F1: 0.9832; Recall: 100%; Latency: 0.05 ms.**

Reconciling PR-AUC Ranking vs. Operational Deployment: A crucial scientific insight revealed by our benchmark is the dual perspective between theoretical mathematical ranking and operational railway dispatching:

- **Theoretical Mathematical Ranking (PR-AUC):** Unsupervised models—**VAE** and **LSTM Autoencoder**—rank #1 and #2 with a near-perfect **0.9833 PR-AUC**. Because continuous reconstruction error (${MSE}(x, hat{x})$) naturally separates healthy vs. abnormal sequences across all potential decision thresholds from 0 to $ ∈ fty$, the integrated area under the curve is exceptionally high. However, at a standard default decision threshold ($τ = 0.50$), their uncalibrated error magnitude yields low precision (15.89% for LSTM-AE, 33.10% for VAE), which would translate to 67% to 84% false alarms in a real depot.
- **Operational Railway Deployment (F1-Score & Precision):** For operational train dispatching where false alarms cause expensive unnecessary depot withdrawals, supervised hybrid models are superior. **Temporal Fusion Transformer (TFT)** achieves **0.9817 PR-AUC, 99.95% Recall, and 96.74% Precision (F1: 0.9832)**, while **CNN-BiLSTM with Attention** achieves **100.00% Recall, 96.66% Precision** with an edge latency of only **0.05 milliseconds**.

VI. Conclusion and Future Work

(a) Conclusion

This project demonstrates the feasibility, effectiveness, and operational viability of deep sequential architectures for predictive maintenance of Air Production Units in passenger metro trains. By analyzing over 1.5 million telemetry readings across 16 physical sensor channels from Metro do Porto, we established an end-to-end, leak-free pipeline with strict chronological partitioning and training-only feature standardization.

Benchmarking across 10 deep learning architectures revealed that the **Variational Autoencoder (VAE)** and **LSTM Autoencoder** excel at theoretical anomaly separation across the complete threshold spectrum (PR-AUC: 0.9833), while the **Temporal Fusion Transformer (TFT)** and **CNN-BiLSTM with Attention** deliver outstanding operational reliability, achieving 100% recall with over 96.7% precision and sub-millisecond edge execution latencies. Furthermore, our Dual-Evaluation framework and unsupervised validation threshold calibration successfully resolved cross-dataset transfer collapse, demonstrating how physical railway telemetry can be transformed into reliable, early diagnostic intelligence without operational disruption.

(b) Future Work

1. **Onboard Edge Hardware Deployment:** Quantize the trained PyTorch models (INT8 / FP16 precision via TensorRT and ONNX Runtime) and deploy them onto an NVIDIA Jetson Orin Nano embedded edge controller to evaluate real-time watt-per-inference efficiency.
2. **Remaining Useful Life (RUL) Prognostics:** Transition from binary pre-warning classification to continuous RUL regression, estimating the remaining operational hours before compressor pressure seals degrade below safety thresholds.
3. **Multi-Modal Fleet Telemetry Fusion:** Incorporate high-frequency acoustic emissions and vibration accelerometry alongside thermodynamic pressure telemetry to enable localized multi-component fault isolation across bearings, dryer towers, and motor windings.
4. **Automated Self-Supervised Threshold Calibration:** Develop continuous online threshold adaptation algorithms using extreme value theory (EVT) to dynamically update alarm thresholds as ambient seasonal temperatures shift.
5. **Explainable AI & Operator Decision Support:** Integrate SHAP (SHapley Additive exPlanations) and multi-head attention weight visualizers into depot dashboard interfaces, presenting maintenance technicians with interpretable root-cause sensor attributions alongside automated work orders.
6. **Federated Multi-Transit Learning:** Implement privacy-preserving federated learning protocols enabling multiple metro transit authorities (e.g., Porto, Vienna, Madrid) to collaboratively train generalized predictive models without sharing proprietary raw telemetry.

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